

These are notes I'm taking for Ben Hayes's Ergodic Theory class for Fall 2026.

The main texts are Petersen's *Ergodic Theory* and Walters's *An Introduction to Ergodic Theory*, as well as some other

General Motivation and Common Examples

The basic structure of ergodic theory is a standard probability space (X, μ) and, in its most basic form, a measurable map $T: X \rightarrow X$ satisfying $T_*\mu = \mu$. Recall that the pushforward measure is defined by

$$T_*\mu(A) = \mu(T^{-1}(A)).$$

We call T a measure-preserving transformation.

Generalizing this, if T is invertible, then we may take a group action $G \curvearrowright (X, \mu)$ by measure-preserving transformations. If T is not invertible, then we may consider a semigroup action $N \curvearrowright (X, \mu)$.

The main items we are interested in are analytical considerations, such as

- asymptotic structure
- connections to unitary representation theory
- isomorphism.

Ergodic theory appears in a number of fields, such as

- unitary representation theory
- analytic and geometric group theory
- functional analysis
- probability theory
- geometry
- analytic number theory.

Background

More aptly, this may be written as “background.” Most of these results are covered in a functional analysis class.

Definition: A measurable space (X, Σ) is called *standard Borel* if there is a separable and completely metrizable space (also known as a Polish space) K such that $(K, \mathcal{B}) \cong (X, \Sigma)$ are isomorphic as measurable spaces, where $\mathcal{B}$ denotes the Borel σ -algebra on K .

Remark: In the special case of standard probability spaces, we may take K to be compact. This emerges from the fact we are only concerned with Radon measures.

Example: The following are examples:

- finite or countably infinite sets;
- products of standard Borel spaces;
- $[0, 1] \cong S^1$;
- $(0, 1) \cong \mathbb{R} \cong [0, 1] \times \mathbb{Z}$;
- $\mathbb{R}^{\mathbb{N}}$, where we use the complete distance metric

$$d(t, s) = \sum_{j=1}^{\infty} 2^{-j} \frac{|t_j - s_j|}{1 + |t_j - s_j|}$$

to provide the desired isomorphism;

- the set $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ subset of the Polish space $\mathbb{R}$, so it is also a Polish space.

Theorem: If (X, Σ) is standard Borel and B is Borel, then $(B, \Sigma|_B)$ is standard Borel.

Theorem (Riesz Representation Theorem): Let K be a compact Hausdorff space, and $L: C(K) \rightarrow \mathbb{C}$ a continuous linear functional. Then, there is a unique complex-valued Borel measure such that

$$L(f) = \int f d\mu.$$

Furthermore, this measure is a Radon measure.

Remark: From the Riesz Representation Theorem, it can be shown that the following hold:

- L is positive if and only if μ is a positive measure;
- $L(1) = 1$ if and only if μ is a probability measure.

One of the fundamental constructions in ergodic theory, especially when we are concerned with actions by a countable discrete group on a standard probability space, is the projective limit of a (countable) family of standard probability spaces.

Consider a system of standard probability spaces $(X_k, \Sigma_k)_{k=1}^\infty$ with connecting maps $\varphi_k: X_{k+1} \rightarrow X_k$ satisfying $(\varphi_k)_* \mu_{k+1} = \mu_k$.

Then, the projective limit

$$\begin{aligned} X_\infty &= \varprojlim X_k \\ &= \left\{ (x_k)_k \in \prod_{k=1}^\infty X_k : \varphi_k(x_{k+1}) = x_k \right\}. \end{aligned}$$